

Internal Bank Policies & Lending Standards

Governing the BFSI AI-Driven Loan Decisioning Workflow

Document No: BP-LEND-2025-01 | Version: 1.0 | Effective: 01 May 2026

Scope: This document defines the internal bank lending policies, thresholds, and operational rules that are enforced by each node in the LangGraph-based decisioning workflow. These policies are calibrated to comply with RBI regulations while reflecting the bank's specific risk appetite and product strategy.

Table of Contents

#	Policy Section	Page (approx.)
1	Data Handling & PII Masking Policy	3
2	Credit Score Band & Limit Policy	4
3	Public Record Severity & Hard-Decline Policy	5
4	Credit Utilisation Adjustment Policy	6
5	Debt Exposure & Leverage Policy	7
6	Inquiry Velocity & Fraud Control Policy	8
7	Payment Behaviour Scoring Policy	9
8	Income Verification & DTI Policy	10
9	RAG Knowledge Retrieval Policy	11
10	Risk Aggregation Weights & Tier Policy	12
11	Lending Decision & Interest Rate Policy	13
12	Counter-Offer Generation Policy	14
13	Final Response & LOS Integration Policy	15
14	Model Fallback & Override Policy	16

1. Data Handling & PII Masking Policy

Applicable Node: `pi_deletion_node` | Entry point to all workflows

All bureau payloads entering the decisioning pipeline must be sanitised before any model processing. The `pi_deletion_node` performs this sanitisation as the mandatory first step. Downstream nodes must only consume the masked payload.

1.1 Masking Rules

Data Category	Fields Removed	Method
Consumer Names	firstName, middleName, surname	Set to null / remove key
Address Components	streetPrefix, streetName, streetSuffix, unitType, unitId, city, state, zipCode	Set to null / remove key
Employer Identity	name (employment tradeline)	Set to null / remove key
Account Numbers	accountNumber (all tradelines)	Set to null / remove key
Court Records	courtName, referenceNumber (public records)	Set to null / remove key

- `raw_experian_data` must never be passed beyond the `pi_deletion_node`.
- `is_pii_masked` flag must be True before any analyzer reads bureau data.
- A deep copy must be created; in-place mutation of the original payload is prohibited.

2. Credit Score Band & Limit Policy

Applicable Node: `credit_score_node`

The bank classifies applicants into four score bands. Each band carries a maximum base lending limit, a risk flag, and a fixed weight (0.25) in the composite risk score. Thresholds below represent policy defaults; the Indian workflow variant fetches updated thresholds from the RAG knowledge base at runtime.

Band	Score Range (CIBIL/Experian)	Base Lending Limit	Risk Flag	Aggregation Weight
PRIME	750 – 900	■75,00,000 / \$75,000	LOW	0.25
NEAR_PRIME	700 – 749	■50,00,000 / \$50,000	MODERATE	0.25
FAIR	650 – 699	■35,00,000 / \$35,000	MODERATE	0.25
SUBPRIME	< 650	■20,00,000 / \$20,000	HIGH	0.25

- Base lending limit is the ceiling before any adjustment factors are applied.
- Score is extracted from `riskModel[0].score` in the masked Experian payload.
- LLM response type must be logged: 'RAG' (policy-guided) or 'FALLBACK' (defaults above).

3. Public Record Severity & Hard-Decline Policy

Applicable Node: public_record_node

The bank evaluates bankruptcies, liens, and court judgments as credit quality indicators. Severity drives both an adjustment factor (reducing lending capacity) and, in the most severe cases, an immediate hard decline that supersedes all other positive signals.

Severity	Conditions	Adjustment Factor	Hard Decline	Audit Action
NONE	No public records on file	1.00	No	Log NONE severity
LOW	Non-bankruptcy records (liens/judgments only)	0.90	No	Log LOW; note record type
MODERATE	Discharged bankruptcy ≥ 5 years ago	0.75	No	Log MODERATE; note discharge date
SEVERE	Bankruptcy < 5 yrs OR multiple judgments	0.50	Yes	Log SEVERE; trigger hard_decline_flag

3.1 Hard-Decline Trigger Conditions

- Bankruptcy filed within the past **24 months** → hard_decline_flag = True.
- Severity = SEVERE → hard_decline_flag = True.
- Multiple court judgments (≥ 3 in last 3 years) → hard_decline_flag = True.

WARNING: Hard decline flag cannot be overridden by the decision node, counter-offer node, or any downstream process. It represents a policy-level veto.

4. Credit Utilisation Adjustment Policy

Applicable Node: utilization_node

Revolving credit utilisation measures the proportion of available revolving credit currently in use. High utilisation is a leading indicator of financial stress and reduces the bank's willingness to extend additional credit.

Utilisation Range	Risk Level	Adjustment Factor	Node Constant	Rationale
0% – 30%	EXCELLENT	0.95 – 1.00	EXCELLENT_ADJ	Low revolving debt; healthy cash flow
31% – 50%	GOOD	0.80 – 0.95	GOOD_ADJ	Moderate usage; manageable
51% – 75%	MODERATE	0.65 – 0.80	MODERATE_ADJ	Elevated; monitor closely
> 75%	CRITICAL	0.50 – 0.65	CRITICAL_ADJ	Near-maxed; high stress

- Only accounts with *revolvingOrInstallment* = 'R' are included in the calculation.
- Adjustment factor multiplies the credit-score-derived base limit.
- $\text{utilisation_ratio} = \text{total_balance} \div \text{total_credit_limit}$ (revolving accounts only).

5. Debt Exposure & Leverage Policy

Applicable Node: debt_exposure_node

Total outstanding debt across all open accounts is assessed to ensure the applicant's leverage does not exceed the bank's risk tolerance thresholds.

Exposure Risk	Debt-to-Income Proxy	Monthly Obligation Ceiling	Impact on Decision
LOW	< 2× gross annual income	< 15% of monthly income	Full capacity available
MODERATE	2× – 4× gross annual	15% – 30% of monthly	Moderate capacity reduction
HIGH	4× – 6× gross annual	30% – 45% of monthly	Significant reduction
EXTREME	> 6× gross annual income	> 45% of monthly income	Near-decline; counter-offer likely

- Only open tradelines (openOrClosed = 'O') are included.
- Monthly obligations from this node feed the income node DTI calculation.

6. Inquiry Velocity & Fraud Control Policy

Applicable Node: inquiry_node

Credit inquiry velocity over a rolling 12-month window is monitored to detect financial distress, credit shopping, or potential fraud. A penalty factor reduces the final approved amount proportionally.

Inquiry Count (12 months)	Velocity Risk	Penalty Factor	Action Required
0 – 1	LOW	1.00 (no reduction)	Normal processing
2 – 3	MODERATE	0.90	Document inquiry sources in file
4 – 6	HIGH	0.70	Enhanced due diligence; flag for review
≥ 7	CRITICAL	0.50	Fraud screening; possible referral to risk team

- Penalty factor is applied multiplicatively in the decision node's capacity calculation.
- Inquiry count is restricted to hard inquiries (credit-seeking) only; soft inquiries are excluded.
- Applications with ≥ 3 inquiries from the same lender within 30 days trigger a fraud alert flag.

7. Payment Behaviour Scoring Policy

Applicable Node: `payment_behavior_node`

Payment history is the single most predictive indicator of future default. The bank scores applicants on a 0–100 behaviour scale based on delinquency counts and chargeoff history extracted from all tradelines.

Risk Band	30+ DPD Count	Chargeoff Present	Behaviour Score	Aggregation Sub-Score
EXCELLENT	0	No	90 – 100	90
GOOD	1 – 2	No	75 – 89	80
FAIR	3 – 5	Old (> 2 yrs)	50 – 74	65
POOR	6+	Recent (< 2 yrs)	25 – 49	30
UNACCEPTABLE	Multiple chargeoffs	Active collections	5 – 24	5

- Behaviour score (0–100) is directly used in the risk aggregator.
- `chargeoff_history` = True if any tradeline has been written off.
- UNACCEPTABLE rating with active collections may independently trigger DECLINE in the decision node.

8. Income Verification & DTI Policy

Applicable Node: `income_analysis_node`

The bank enforces strict Debt-to-Income (DTI) limits to ensure all approved loans are within the borrower's verified repayment capacity.

DTI Range	Income Risk	Affordability Flag	Policy Action
0% – 30%	LOW	True	Approve full requested amount (capacity permitting)
31% – 40%	MODERATE	True	Approve with standard documentation
41% – 50%	HIGH	True	Approve but flag for post-disbursement monitoring
> 50%	UNACCEPTABLE	False	Immediate decline; counter-offer with reduced principal

DTI = (Sum of all open monthly obligations + new EMI) ÷ verified gross monthly income.

Income must be verified via bank statement summary (monthly_income field). If income is unverifiable, DTI defaults to UNACCEPTABLE and the application is declined.

- Maximum allowable DTI for counter-offer options: 40%.
- Monthly income source: bank_statement_summary.monthly_income field.

9. RAG Knowledge Retrieval Policy

Applicable Node: `rag_retrieval_node` (Indian variant only)

The Indian workflow variant retrieves policy thresholds and guidance at runtime from a Qdrant vector database, enabling dynamic policy updates without code changes. This section governs how RAG results are used.

9.1 Per-Node Retrieval Strategy

Node	Concern Query Sent to Qdrant	Top-K
credit_score	score band thresholds, base limits, risk flags	3
public_record	bankruptcy severity, adjustment factors, hard-decline rules	3
utilization	revolving balance thresholds, adjustment factors	3
exposure	debt leverage limits, monthly obligation caps	3
inquiry	inquiry velocity thresholds, penalty structures	3
behavior	delinquency severity, chargeoff rules	3
income	DTI thresholds, affordability rules	3

9.2 Fallback Policy

- If RAG retrieval returns empty for a node, that node must use hardcoded fallback defaults (documented in Sections 2–8 above).
- `llm_response_type` field must be set to 'FALLBACK' when defaults are used.
- RAG results from multiple collections are deduplicated by chunk ID before being forwarded to nodes.
- Policy documents in Qdrant must be re-indexed whenever RBI issues a new circular or the bank updates its internal policy.

10. Risk Aggregation Weights & Tier Policy

Applicable Node: `risk_aggregator_node`

The risk aggregator combines seven independent signals using a fixed weighted average. Weights reflect the bank's historical loss attribution analysis and may only be modified following a formal Model Governance review.

10.1 Factor Weights

Risk Factor	Weight	Node Source	Sub-Score Normalisation (text → 0-100)
Credit Score	25%	<code>credit_score_node</code>	Score band → (score-300)/5.5 clamped 0-100
Payment Behaviour	15%	<code>behavior_node</code>	Uses raw behaviour_score (already 0-100)
Public Records	15%	<code>public_record_node</code>	NONE=100, LOW=90, MODERATE=60, SEVERE=30
Utilisation	15%	<code>utilization_node</code>	EXCELLENT=90, GOOD=60, MODERATE=65, CRITICAL=10
Income / DTI	15%	<code>income_node</code>	LOW=90, MODERATE=60, HIGH=65, UNACCEPTABLE=5
Debt Exposure	10%	<code>exposure_node</code>	LOW=90, MODERATE=60, HIGH=65, EXTREME=5
Inquiry Velocity	5%	<code>inquiry_node</code>	LOW=90, MODERATE=60, HIGH=65, CRITICAL=10

10.2 Risk Tier Bands

Aggregate Score (0-100)	Tier	Decision Disposition	Target Interest Range
80 – 100	A	APPROVE (full amount)	7.5% – 9.0%
65 – 79	B	APPROVE (standard terms)	10.0% – 12.5%
50 – 64	C	COUNTER_OFFER likely	13.5% – 16.0%
35 – 49	D	COUNTER_OFFER or DECLINE	18.0% – 22.0%
< 35	F	DECLINE (hard)	N/A

WARNING: If `hard_decline_flag = True`, `aggregated_risk_score` is forced to 0.0 and `Tier = F` regardless of other signals. This is a non-negotiable policy veto.

11. Lending Decision & Interest Rate Policy

Applicable Node: decision_llm_node

The decision node applies a structured decision protocol based on the aggregated risk score, individual adjustment factors, and income affordability results.

11.1 Decision Protocol

Step	Condition	Action
1	Tier = F OR hard_decline_flag = True	DECLINE immediately
2	Assign interest rate per tier (table 11.2)	Set interest_rate
3	Compute max_approved = base × adj_factors	Capacity ceiling established
4	affordability_flag = False	DECLINE (income constraint)
5	Requested ≤ max_approved	APPROVE at requested amount
6	Requested > max_approved AND max_approved > 0	COUNTER_OFFER
7	max_approved = 0	DECLINE
8	Disbursement = approved × 0.98	2% origination fee deducted

11.2 Interest Rate Schedule by Risk Tier

Tier	Annual Rate	Monthly Rate (approx.)	Product Application
A	7.5%	0.625%	Personal loans, home loans, MSME term loans
B	10.0%	0.833%	Personal loans, vehicle finance
C	13.5%	1.125%	Unsecured personal credit, top-up loans
D	18.0%	1.500%	Micro-credit, secured personal loans only
F	N/A	N/A	No lending permitted

Capacity Calculation: $\text{max_approved} = \text{credit_score.base_limit_band} \times \text{public_record.adjustment_factor} \times \text{utilization.adjustment_factor} \times \text{inquiry.penalty_factor}$

12. Counter-Offer Generation Policy

Applicable Node: counter_offer_node

When the requested amount exceeds the applicant's assessed capacity, the counter-offer node generates restructured alternatives. At least two viable options must be presented.

Option ID	Description	Principal Adjustment	Tenure Adjustment	Min. Tenure
OPT_REDUCED_AMOUNT	Reduce principal, keep original term	~15% reduction ($\times 0.85$)	No change	12 months
OPT_EXTENDED_TERM	Reduce principal + extend term	~30% reduction ($\times 0.70$)	+12 months	36 months

12.1 Affordability Ceiling Rules

Max EMI = verified monthly income $\times$ 35%

All counter-offer options must produce an EMI $\leq$ this ceiling. Origination fee of 2% applies to disbursement on all options.

12.2 Counter-Offer Trigger

- Counter-offer node executes ONLY when decision_result.decision = 'COUNTER_OFFER'.
- counter_offer_logic must explain why the original amount was not sanctioned.
- original_request_dti (the DTI that triggered counter-offer) must be logged.
- If no viable option can be structured (max_approved = 0), the node passes control to DECLINE.

13. Final Response & LOS Integration Policy

Applicable Node: final_response_node

The final_response_node composes the LOS-compatible JSON payload. The payload schema is standardised across all three decision outcomes (APPROVE / COUNTER_OFFER / DECLINE) to facilitate LOS integration.

13.1 Mandatory Fields by Decision Type

Field	APPROVE	COUNTER_OFFER	DECLINE
application_id	✓	✓	✓
correlation_id	✓	✓	✓
decision	✓	✓	✓
risk_tier	✓	✓	✓
risk_score	✓	✓	✓
loan_details.*	✓	–	–
counter_offer.*	–	✓	–
decline_reason	–	–	✓
reasoning_steps	–	–	✓
rag_response (Indian only)	✓	✓	✓

- Disbursement amount (after 2% fee) must always be included in loan_details.
- Indian variant must attach the full rag_pool and per-node context in rag_response.

14. Model Fallback & Override Policy

Applicable Nodes: All LLM-based nodes

Every LLM-based analyzer node must implement a deterministic fallback path that activates when the LLM call fails, times out, or returns an unparseable response. Fallback decisions must be logged and flagged for human review.

14.1 Fallback Defaults per Node

Node	Fallback Trigger	Fallback Behaviour	Log Action
credit_score_node	LLM unavailable/error	Use hardcoded score band thresholds (Sec 2)	Log FALLBACK flag
public_record_node	LLM unavailable/error	Use hardcoded severity rules (Sec 3)	Log FALLBACK flag
utilization_node	LLM unavailable/error	Use hardcoded utilisation tiers (Sec 4)	Log FALLBACK flag
exposure_node	LLM unavailable/error	Use hardcoded exposure tiers (Sec 5)	Log FALLBACK flag
inquiry_node	LLM unavailable/error	Use hardcoded velocity tiers (Sec 6)	Log FALLBACK flag
behavior_node	LLM unavailable/error	Use hardcoded behaviour rubric (Sec 7)	Log FALLBACK flag
income_node	LLM unavailable/error	Use hardcoded DTI bands (Sec 8)	Log FALLBACK flag
decision_llm_node	LLM unavailable/error	Score $\geq 75 \rightarrow$ APPROVE; 55-74 $\rightarrow$ COUNTER; $< 55 \rightarrow$ DECLINE	Log FALLBACK flag
counter_offer_node	LLM unavailable/error	85%/70% of requested; +12m tenure option	Log FALLBACK flag

14.2 Human Override Rules

- Any decision made via FALLBACK path must be flagged for human credit officer review within 24 hours.
- Human overrides of AI decisions must be logged with officer ID, timestamp, and override reason.
- Override statistics must be included in the monthly model performance report.

WARNING: FALLBACK path decisions carry higher model risk. Accumulation of FALLBACK decisions (> 5% of daily volume) triggers an automatic escalation to the Technology Risk team.